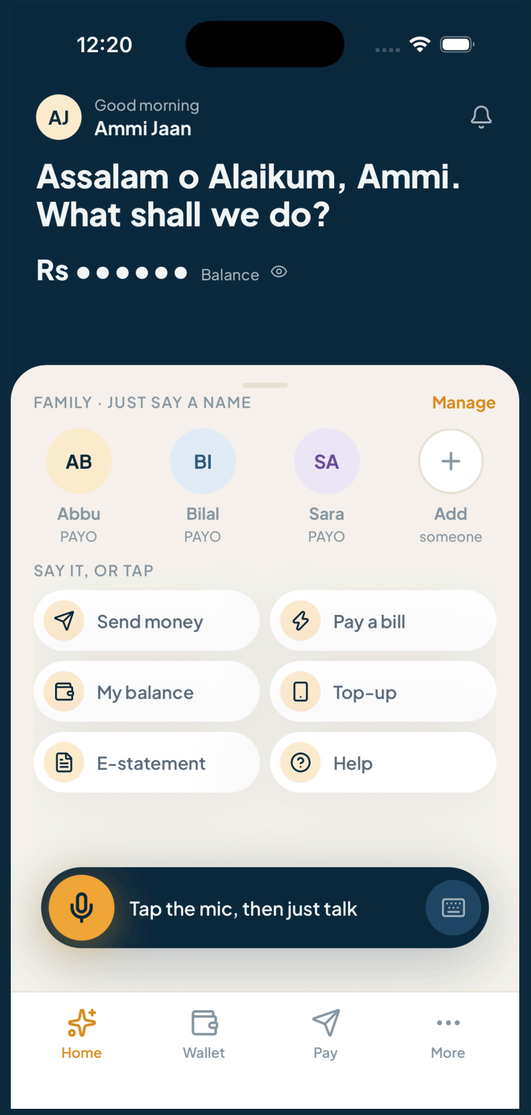
PAYO · HACKATHON 2026 · WORKING DEMO

Banking you can talk to.

Hands-free banking with an AI agent. Say what you want, in English or Urdu. Your PIN does the rest.

48 capabilities | 32 card kinds | English · Urdu · Roman Urdu



THE PROBLEM

Every banking action is a stack of screens, taps and forms.

Menus assume reading

Our parents do not browse a bank. They want to ask one question and get one answer.

Forms assume typing

An IBAN field needs both hands and all your attention. It is where the conversation ends.

Confirmations assume doubt

Under pressure anyone taps yes. A sixth confirmation screen protects nobody.

And the scam callers know it: "your account will be blocked", "you won a prize", "read me the code you just got".

THE IDEA

Say what you want. Your PIN does the rest.

48
tools: one per thing the app can do

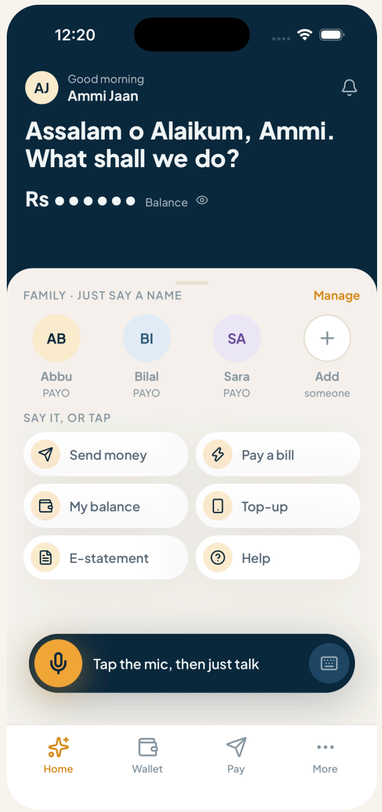
32
card kinds: every answer is a card

3
input languages: English, Urdu, Roman Urdu

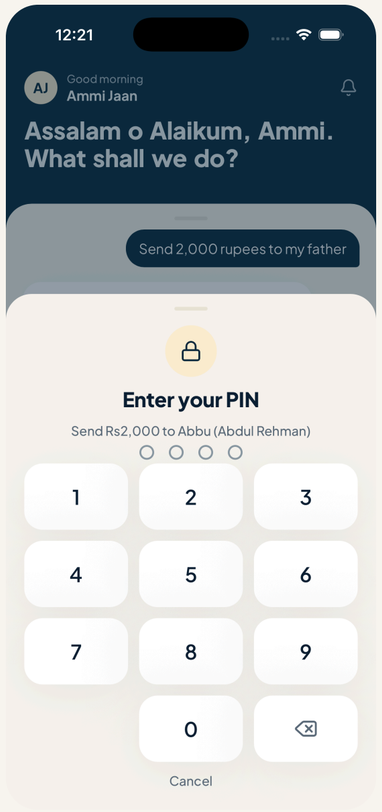
1
safety gate: prepare, PIN, execute once

HERO DEMO 1 · HANDS-FREE

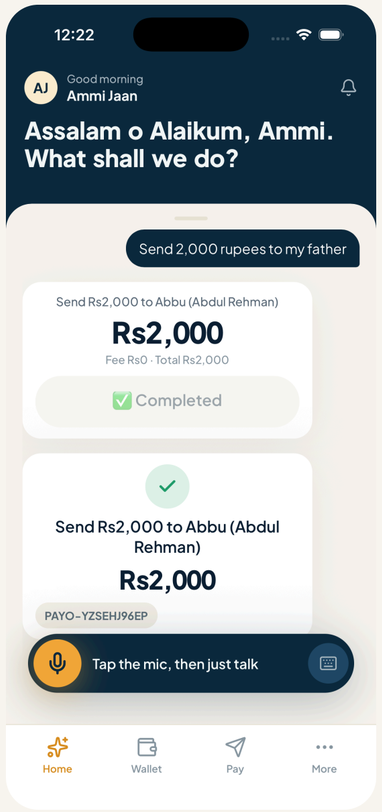
“Send two thousand rupees to my father.”



Family on Home. Tap Abbu, or just say the word.



Straight to the PIN. No bank question, no number, no “shall I send?”



Done, and spoken. The card names him the way she does.

HOW “MY FATHER” BECOMES A PAYMENT

The model never guesses who. The backend does the looking up.

What you say	Who PAYO finds
father · dad · abbu · abba · baba · walid · والد	**Abbu**
mother · mom · ammi · amma · walida · والدہ	**Ammi**
brother · bhai · بھائی · sister · baji · behen	**Bhai · Baji**
بلال (Urdu script) · Bilal · bilal	**Bilal**

Deterministic, not probabilistic

A lexical alias table in the bank core maps 12 relationships, in three scripts, onto saved nicknames. No model in the loop.

Saved people, real titles

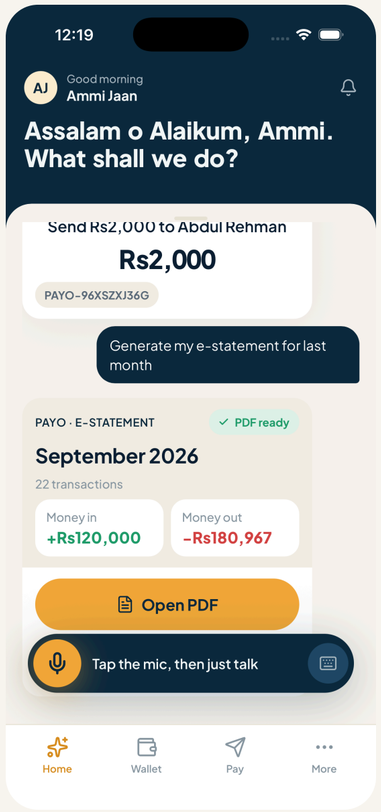
Recipients resolve to an account title before anything moves. An Urdu-script name reaches a PAYO user through their Urdu name.

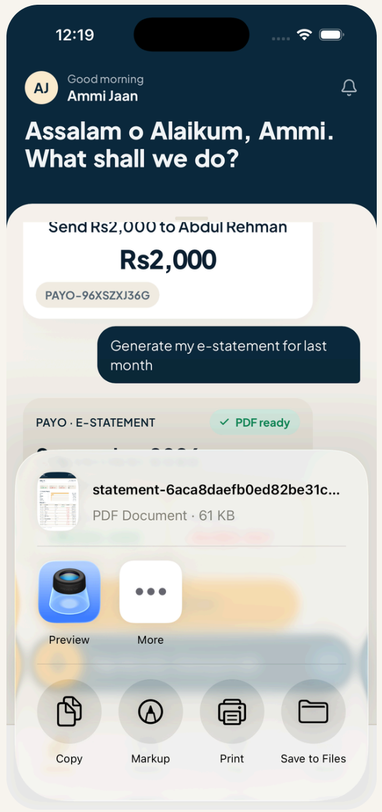
One match plus an amount = the card

Exactly one match and a spoken amount go straight to the confirmation card. Two matches become a question with chips. Never a guess.

HERO DEMO 2 · THE FORM THAT DISAPPEARED

“Generate my e-statement for last month.”





A card that reads like page one

Period, transaction count, money in, money out. PAYO speaks the gist while the card carries the detail.

Open PDF hands over the real file

Built on the spot by the bank core, opened in the share sheet without leaving the conversation. AirDrop it, save it, mail it.

“List my statements” too

Every statement ever generated, as rows with the same Open action. The classic Statements screen is still one tab away.

HANDS-FREE

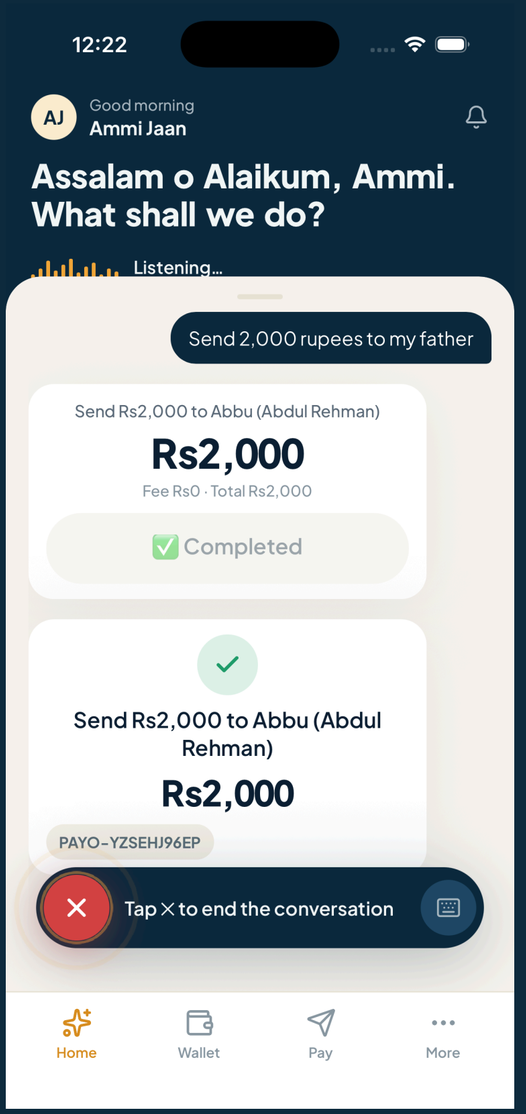
One tap. Then just talk.

Keeps listening after every reply. No tap per turn.

Tap to interrupt while it speaks. Tap the red ✕ to end.

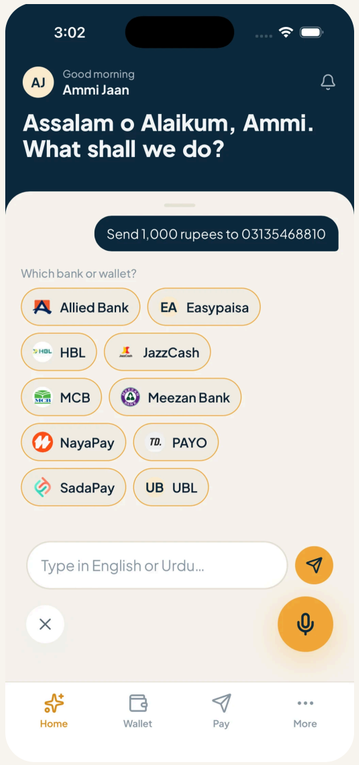
Ends by itself on silence, or after 12 spoken turns.

Cards are spoken, not read. The card carries the detail, the voice carries the gist.

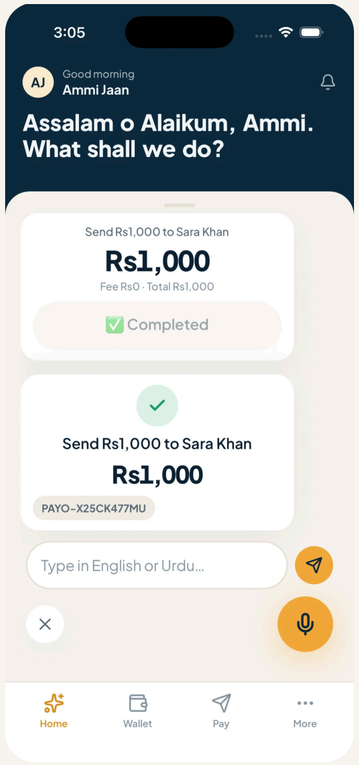


NO MENUS

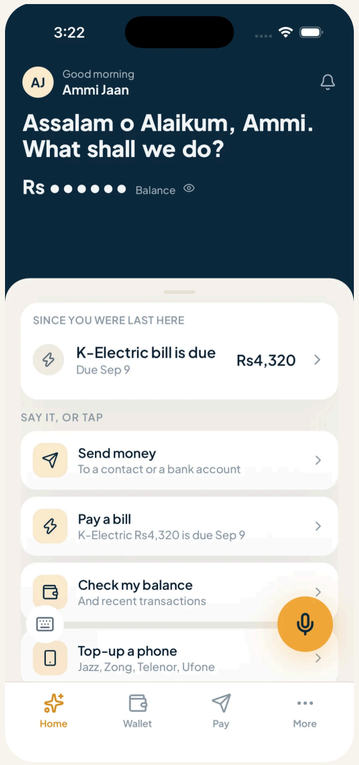
Every answer is a card. Every screen has a spoken equivalent.



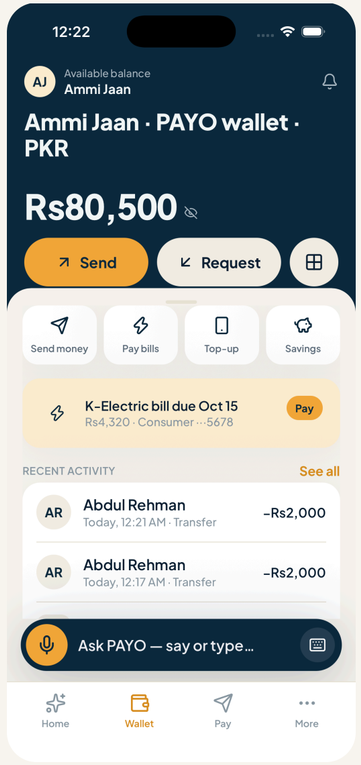
Chips, not an IBAN field



Receipts and spending bars



It speaks first: bills due, money in



Hands still work: the classic wallet

IT STAYS SAFE

The assistant can do everything the app does. It cannot move a single rupee.

Prepare → PIN → execute once

Three steps, one release

A write tool only creates a pending action. The PIN sheet opens inside the conversation; the assistant never sees the PIN. Then one atomic, idempotent, single-use call. Unused actions expire in two minutes.

Guardian · cooling period

A second pair of hands

A trusted contact approves large or unusual sends from their own phone, with their own PIN. People you already pay never wait. Loosening a limit cools off first; tightening is instant.

Scam check-in · age-aware

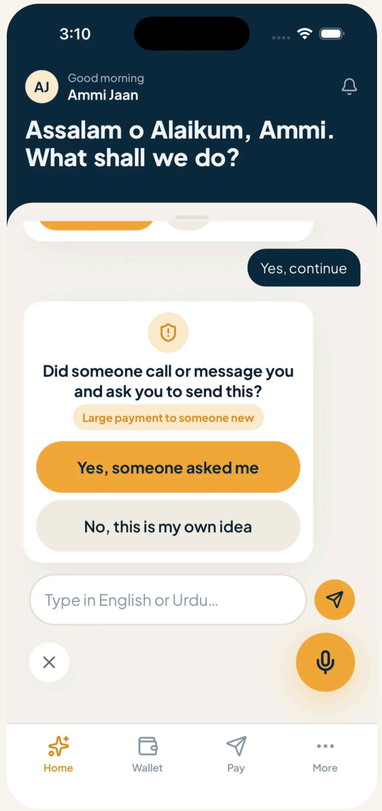
One calm question

“Did someone call or message you and ask you to send this?” A yes cancels the payment and explains, without arguing. Seniors are asked sooner; pressure language asks everyone.

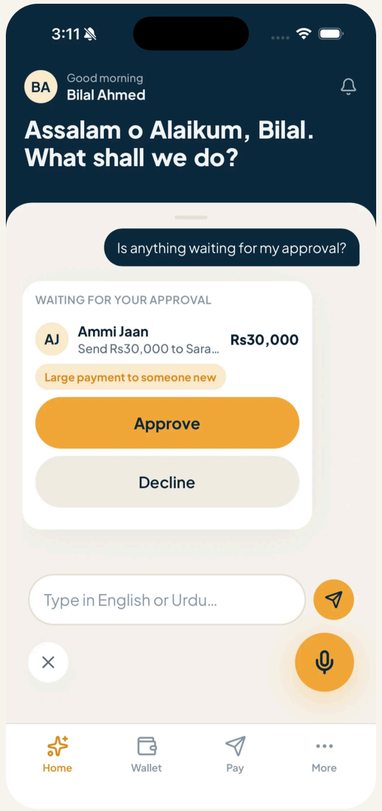
Deterministic guards sit around the model. It may be creative about words, never about actions.

FOR THE FAMILY AROUND THE USER

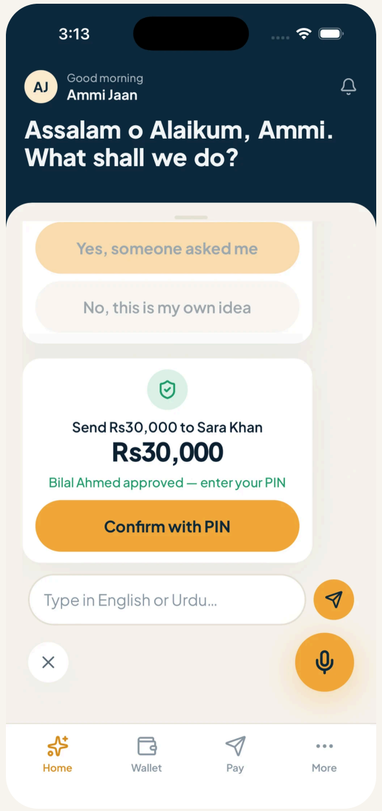
The call every family fears, answered with one question.



The check-in. "Yes, someone asked me" stops it. No lecture.



The guardian decides on his own phone, with his own PIN.



Approved, then her PIN. The sheet opens by itself within seconds.

PAYOFF The check-in call stops.

YOUR LANGUAGE, YOUR PACE

English, Urdu, Roman Urdu. Spoken or typed.

ابو کو دو ہزار روپے بھیجو

Understands all three — “bijli ka bill pay karna hai” works as well as the Urdu script.

Always replies in your language, in Nastaliq, right to left, numbers spoken as words.

“Switch to Urdu” flips the whole app, including the cards already in the conversation.



HOW IT IS BUILT

Three services. The assistant is just another client.

apps/mobile

The phone

Expo SDK 57 · React Native · 32 native card renderers · hands-free loop · PIN sheet · English and Urdu, RTL

→

services/ai

The assistant

FastAPI · LangGraph · Gemini (audio in) · Cartesia (voice out) · 48 tools that call the bank with the user's own token · SSE

→

services/backend

The bank core

Node · Express · MongoDB transactions · pending-action engine · guardian and risk rules · statement PDFs · seeded demo world

Every write the assistant makes is a pending action the bank core still has to run. It holds no database, no PIN, no keys: only the user's token.

HONEST LIMITS

Everything you saw is real. Here is exactly what is not.

Real

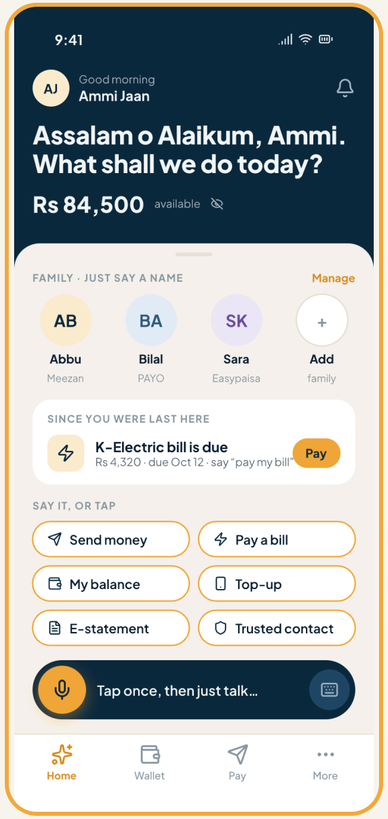
- A working app, three services, one safety gate
- Family by voice, e-statement PDFs, guardian, scam check-in
- 880+ automated tests across the three services
- A live routing gate run against the real model, in both languages
- Design-system UI, Urdu RTL, hands-free loop with cost guards

Not yet

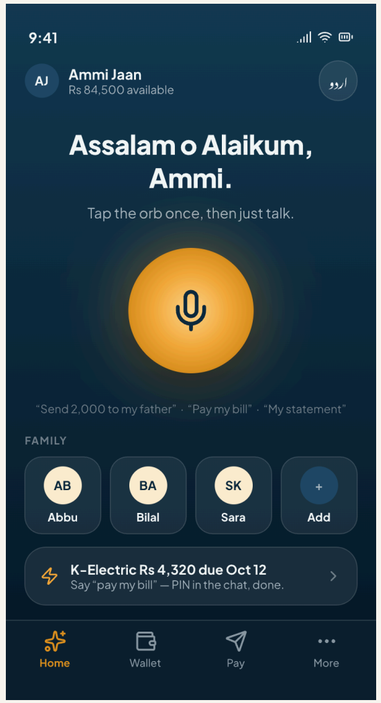
- The world is seeded: balances, titles and bills are deterministic fakes
- Institutions are simulated: no rails, no settlement, no KYC, no real SMS
- Verified on the iOS simulator; Android has not been run in weeks
- Routing stays probabilistic: one intermittent miss per full run
- Statement PDFs are English-only; the app shows Urdu

DESIGN PROCESS

Three directions, explored in Pencil. We shipped A.



A · Navy head, evolved. Family row, chip grid, docked voice bar. Shipped tonight.



B · Voice orb. Immersive, the mic is the hero, glass cards. Next in line.



C · Daylight. Elderly-first: larger type, one giant Talk button, big family tiles.

PAYO

One sentence.
One PIN.
No forms.

Hands-free banking with an AI agent that prepares, never executes. A grandmother who never has to see a form. That is PAYO.

web-flax-beta-22.vercel.app · the 3D story

payo-classic.vercel.app · the site

github.com/munsif12/payo · the code